

Science Communication

Models and elements of science communication

Rui Mata, HS 2026

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Course structure

Session information

Sessions take place Thursdays, 8.15–9.45, Biozentrum, Hörsaal U1.131.

#	Date	Topic	Instructor(s)	Material	Exercise
1	17.09.2026	What is science communication?	Mata	pdf	
2	08.10.2026	Models and elements of science communication	Mata	pdf	
3	15.10.2026	Scientific uncertainty and trust in science	Mata	pdf	
4	22.10.2026	Guidelines for science communication	Mata	pdf	
5	29.10.2026	Science communication in practice	Mata	pdf	
6	05.11.2026	Why do we visualize?	Hil/Lachenmeier		
7	12.11.2026	Structured content	Hil/Lachenmeier		A
8	19.11.2026	Content dimensions and categorization	Hil/Lachenmeier		B
9	26.11.2026	The identity of a dot	Hil/Lachenmeier		C
10	03.12.2026	Networks and relationships	Hil/Lachenmeier		D
11	10.12.2026	Visual language and cultural context & Areas and hierarchies	Hil/Lachenmeier		E
12	17.12.2026	Exam			

Goals for today

- Get an overview of the history, models, and elements of science communication
- Identify stakeholders and audiences (public segmentation) of science communication
- Discuss rationale and practices of evaluation of science communication

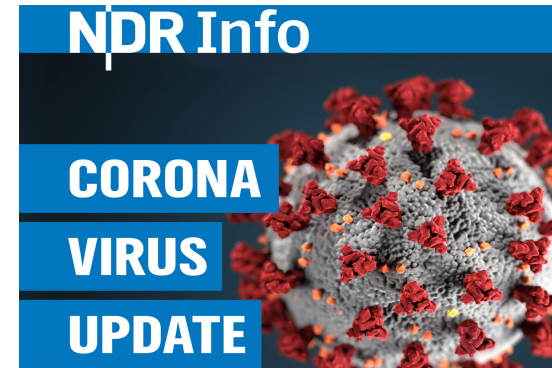
Communicating science: From smallpox to COVID-19



Daniel Bernoulli · 1760
Making risk calculable
Used mathematics and public argument to explain how smallpox inoculation could save lives.



Elvis Presley · 1956
Making vaccination visible
Received a polio vaccination in front of cameras to encourage young people to follow his example.



Corono Virus Update · 2020
Explaining COVID-19
Used podcast conversations to explain emerging COVID-19 research, what scientists knew, and what remained uncertain.

MAJOR CHANGES IN SCIENCE COMMUNICATION OVER TIME?



Historical perspective on science communication

14th -16th centuries

- Rediscovery of classical texts (Greek and Roman), stimulating the revival of natural philosophy and scientific inquiry
- Cabinets of curiosity as informal spaces for discussing the natural world
- Early dissemination of scientific ideas via manuscripts and private letters among scholars

17th -18th centuries

- Ideas of natural philosophy discussed in coffee houses
- Founding of formal institutions to advance science (Royal Society, 1660), introducing the concept of the scientific paper (e.g., *Philosophical Transactions*) and peer review
- Creation of museums (Ashmolean, 1678)

19th century

- Local science societies in England, France, and America fostered wider communication.
- Formal institutions focused on science communication (British Association for the Advancement of Science, 1831)
- Rise of specialized museums (natural history)

Bertemes, J. P., Haan, S., & Hans, D. (Eds.). (2024). *50 essentials on science communication*. De Gruyter.
<https://doi.org/10.1515/9783110763577>

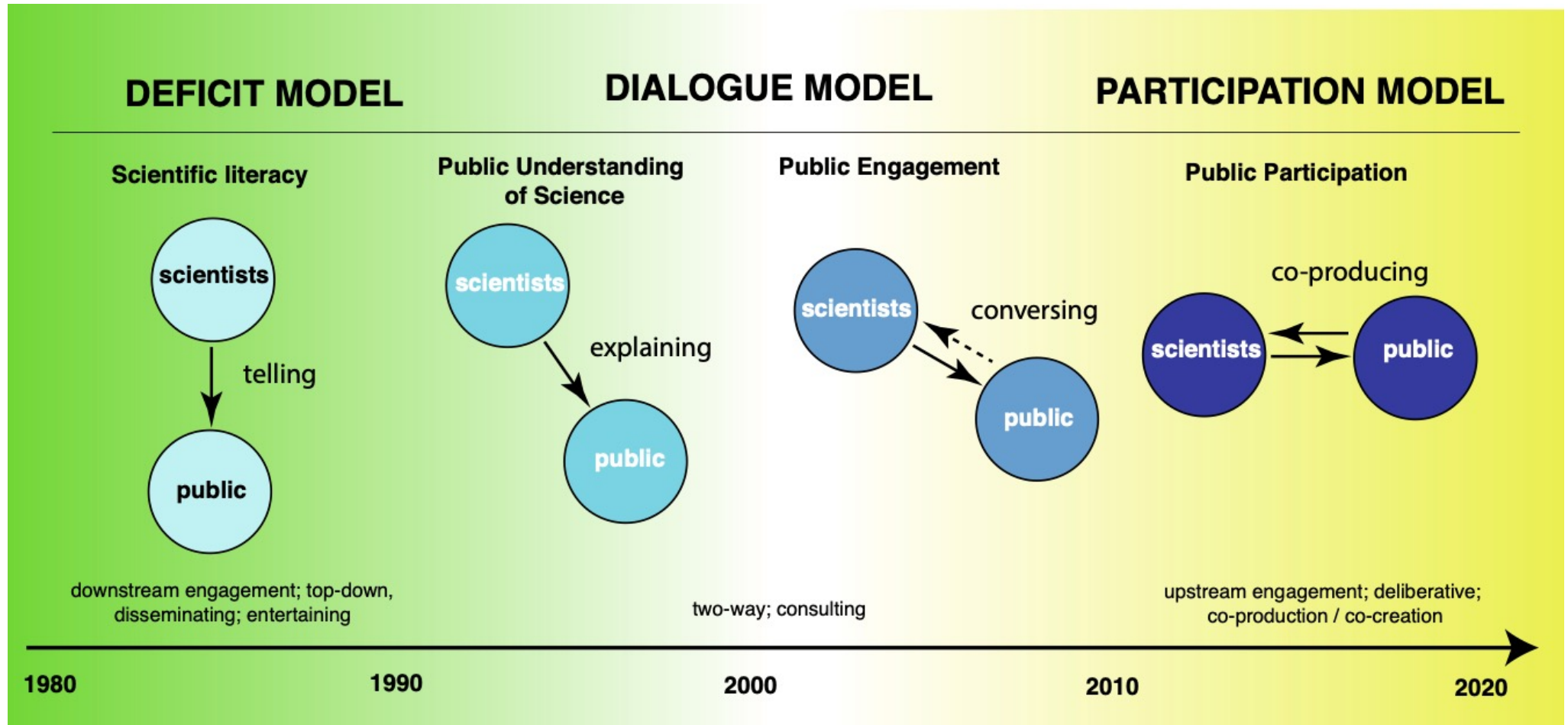
Historical perspective on science communication

20th century until today

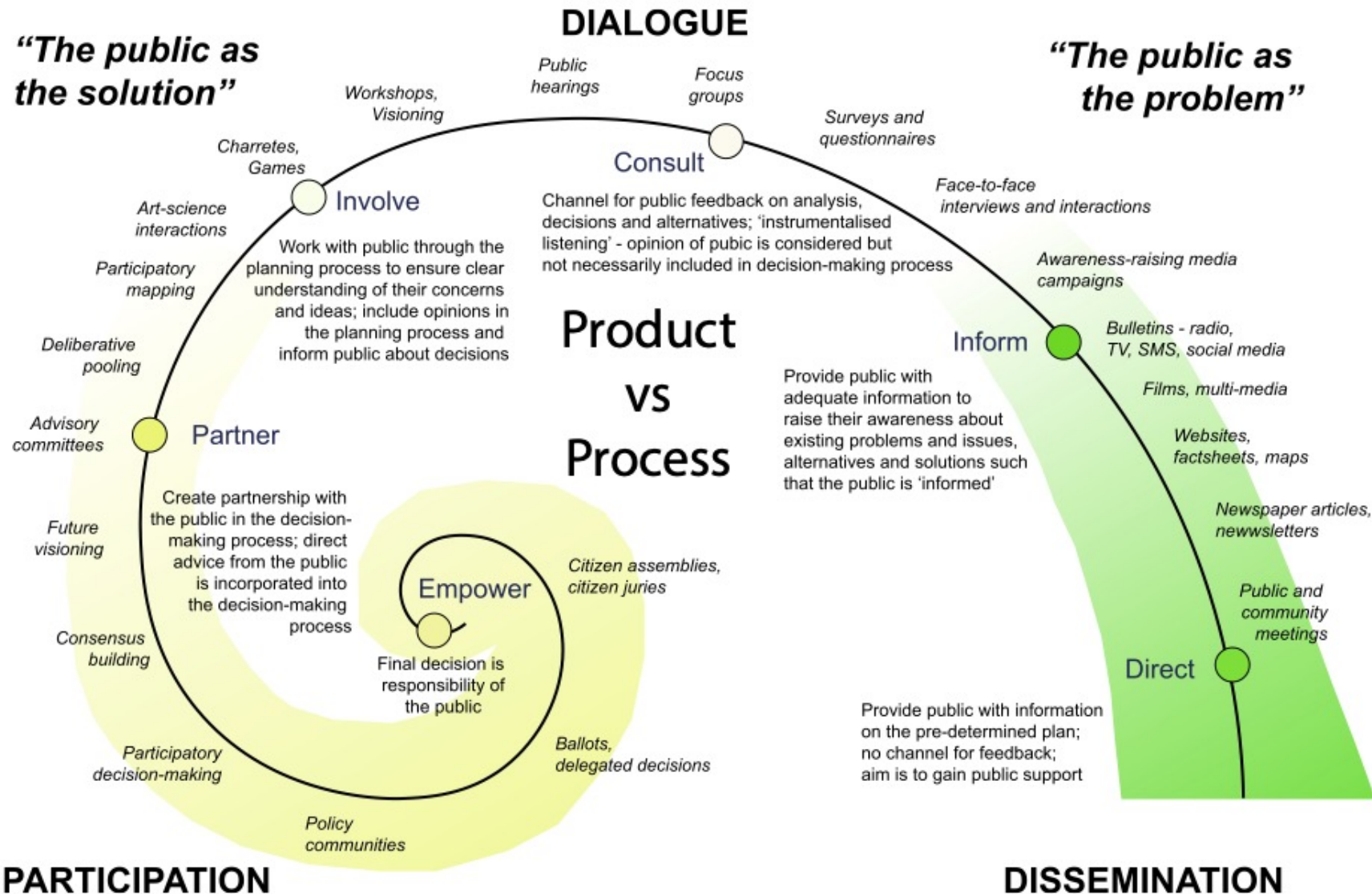
- Rise of mass media and popular science (magazines, radio, public lectures)
- Massive education efforts as part of war (Atomic Energy, 1940s) and public health (vaccination, Polio, 1950s) efforts through films, brochures, and school programs
- Television as major influential medium for science communication (space exploration surrounding Sputnik launch 50s-60s).
- Professionalization of science communication through specific higher-education programs (1970s)
- Growing public concern about environmental issues (Chernobyl, 1980s)
- Rise of the internet (1990s)

21st century

- Social media: many-to-many communication - publics respond, share, organize, and produce content themselves; audiences fragment across channels and platforms; media use patterns diverge sharply between groups



Stewart, I. S. (2024). Advancing disaster risk communications. *Earth-Science Reviews*, 249, 104677.
<https://doi.org/10.1016/j.earscirev.2024.104677>



The ladder direct → inform → consult → involve → partner → empower orders approaches by how much influence the public has over decisions.

Stewart, I. S. (2024). Advancing disaster risk communications. *Earth-Science Reviews*, 249, 104677. <https://doi.org/10.1016/j.earscirev.2024.104677>

- Linear models: science 'speaks truth to power' - knowledge flows one way into policy (cf. deficit).
- Interactive models: iterative exchange between scientists, policymakers, and intermediaries (cf. dialogue).
- Embedded models: knowledge is co-produced with society and policy from the start (cf. participation)

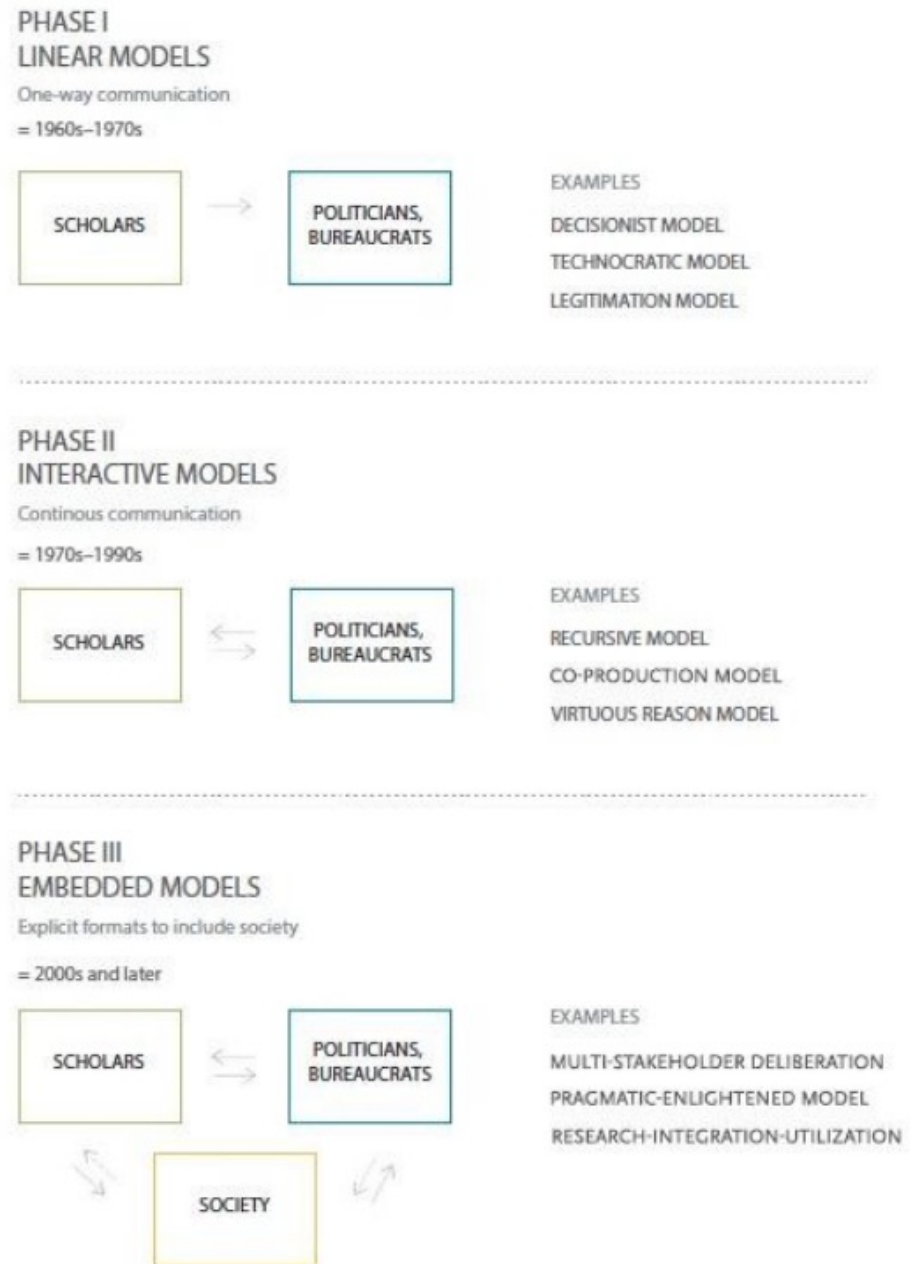


Figure 1. Models of scientific policy advice.

STRENGTHS & WEAKNESSES OF COMMUNICATION MODELS?

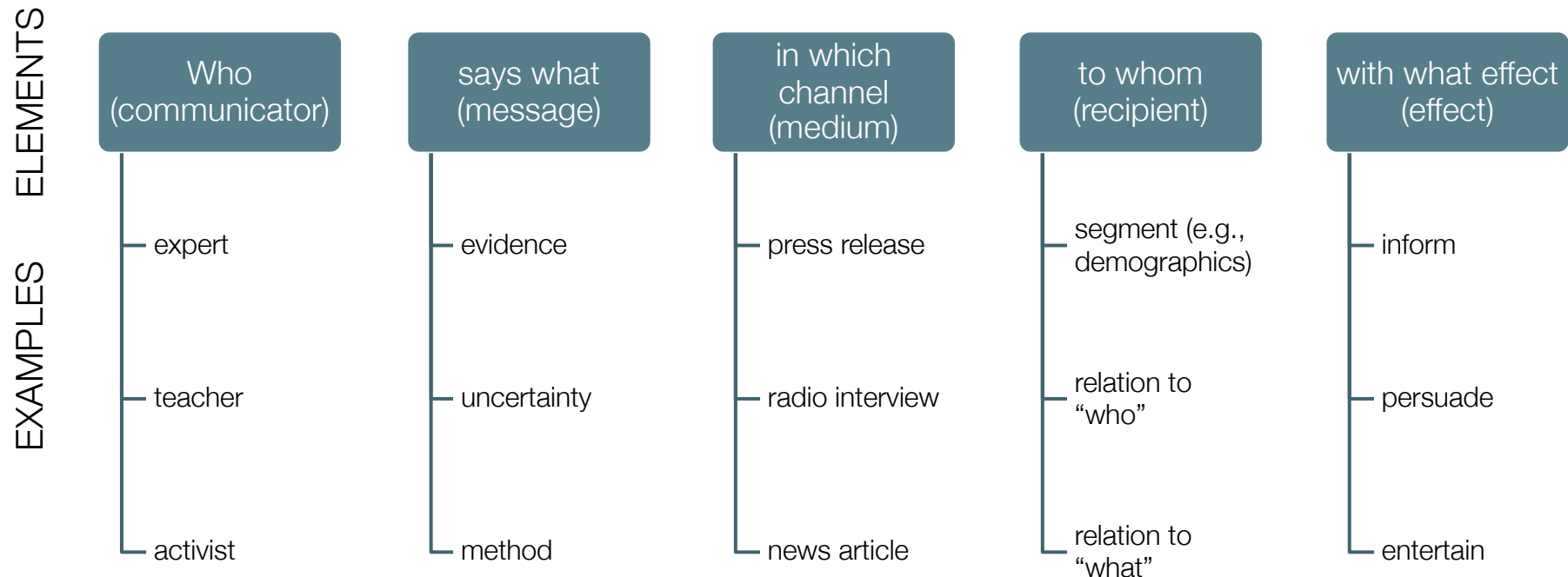


Strengths and Weaknesses of Science Communication Models

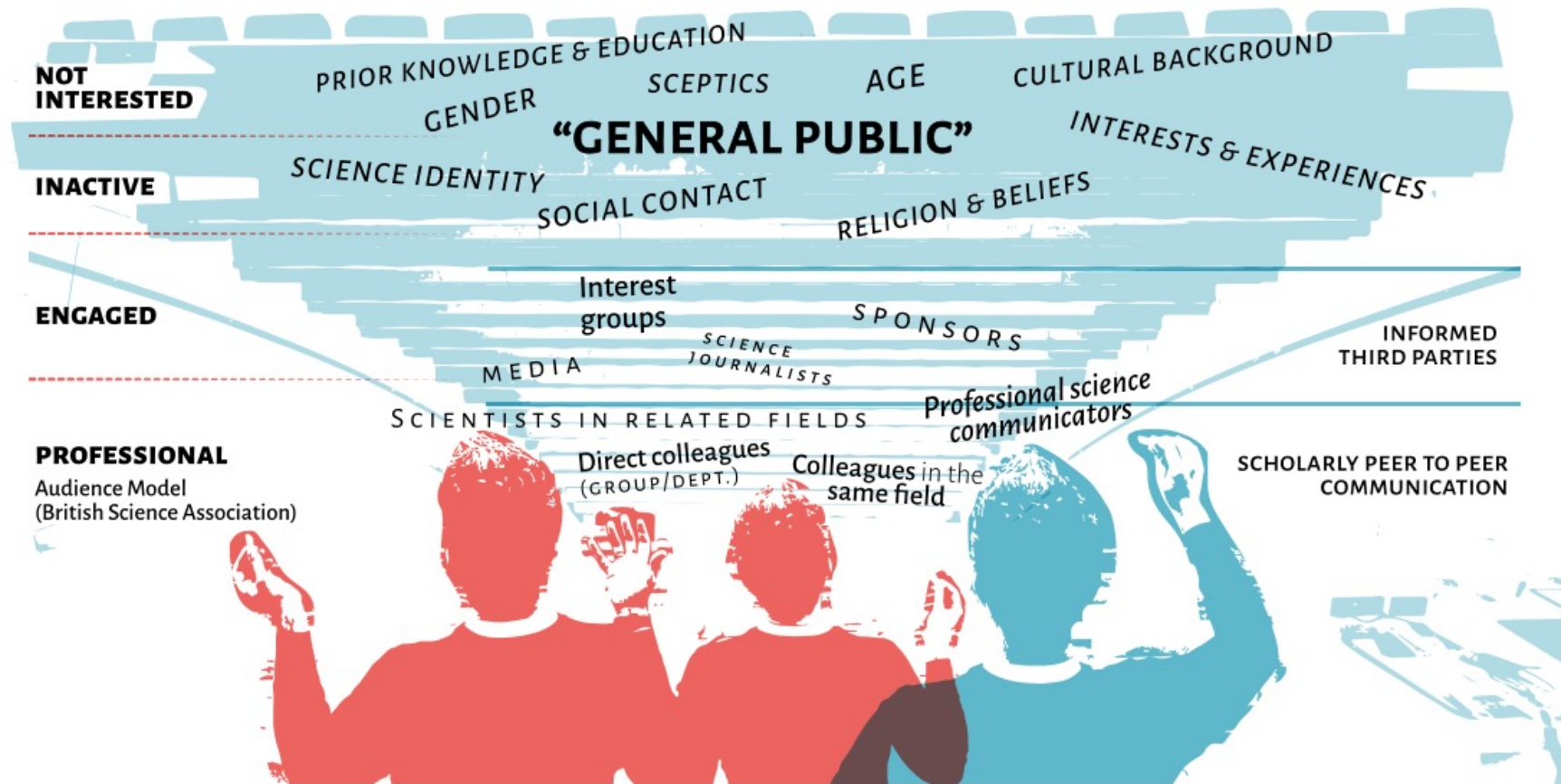
Model	Strengths	Weaknesses	Examples
Deficit Model	Efficient for rapid dissemination; simplifies complex information.	Ignores public values and feedback; one-way communication; limited impact	Public health campaigns (smoking), GMOs and nuclear power debates
Dialogue Model	Builds trust, promotes mutual understanding, and allows tailored communication.	Resource-intensive; power imbalance remains; limited impact and reach;	Public consultations on climate change and renewable energy technologies in the 2000s, resistance to nuclear power
Participation Model	Empowers public; addresses ethical concerns; leads to co-produced, robust solutions.	Resource-heavy, complex; conflicts among stakeholders; scalability issues;	Emergence of citizen science projects (biodiversity monitoring) but challenges in policy consensus (glyphosate debates)

Other (process) models of communication

Lasswell's Model of Communication is a classic framework for analysing the components of communication and it has been applied to science communication to help systematize different **factors** that can be relevant to ensure more effective communication.



Stakeholders



This figure illustrates the Audience Model by the British Science Association, showing the spectrum of stakeholders in science communication, from professional scientists and professional science communicators (e.g., journalists) to the general public with varying levels of interest. Understanding these diverse groups helps tailor communication strategies to better engage each audience and promote effective science communication.

Brokers: Science Journalism

- **Decline of science journalism:** Science journalism started to emerge as a distinct field in the first half of the 20th century. However, since the 1990s, science journalism is facing increasing challenges due to shrinking newsrooms, tighter deadlines, and economic pressures, leading to less specialized and independent coverage. The shift to online media and reliance on press releases further impacts the quality of science reporting, limiting public access to in-depth, reliable information.
- **Fragmentation of science communication:** The decline of science journalism has been accompanied by fragmentation and democratization of science communication. At the same time, there is a larger degree of professionalization and strategic deployment of institutional science communication now being conducted by journalists.



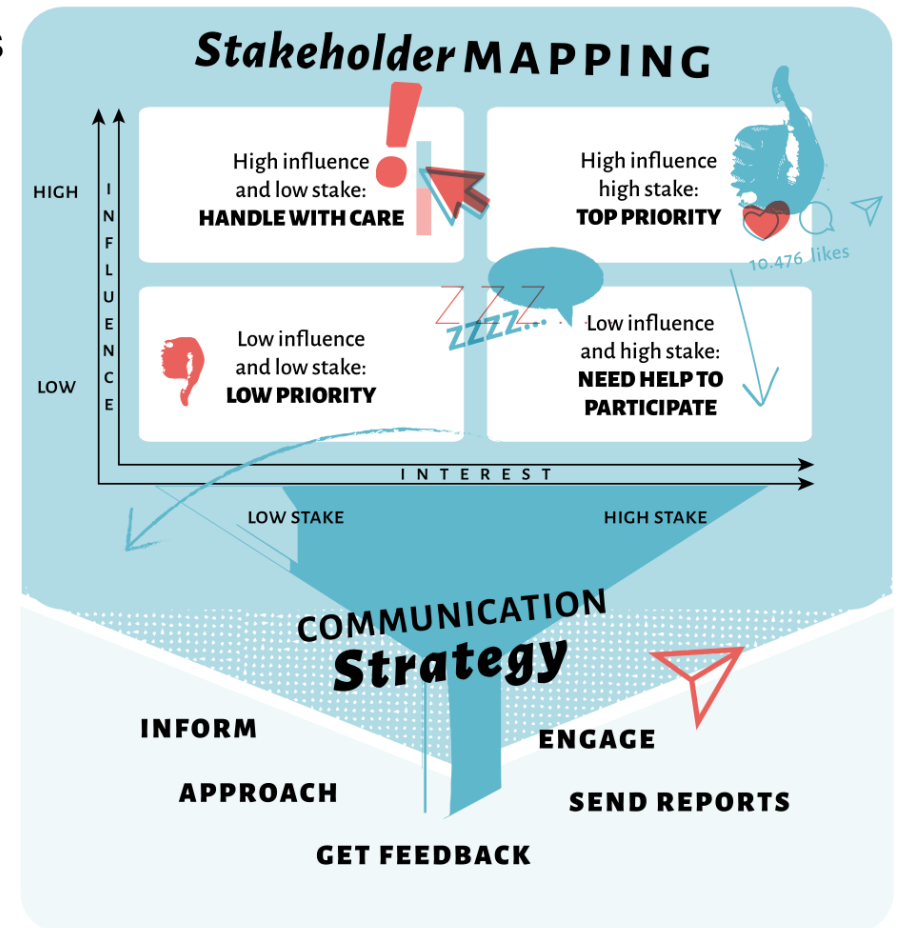
No miracle philanthropy: A project by the Gebert Rüf Foundation and the Mercator Foundation Switzerland aimed to promote science communication in 20 minutes but after funding was discontinued, scientific topics were scaled to a minimum. 15

Schäfer, M. S. (2017). How changing media structures are affecting science news coverage (K. H. Jamieson, D. M. Kahan, & D. A. Scheufele, Eds.; Vol. 1). Oxford University Press. <https://doi.org/10.1093/oxfordhb/9780190497620.013.5>

Stakeholder Mapping and Segmentation

Public segmentation can improve the effectiveness and reach of science communication by acknowledging and addressing the diverse needs of the audience.

- **Tailored messaging:** People vary in their trust, knowledge, and interest in science, so a generalized communication strategy may fall short; segmentation allows communicators to customize messages for specific groups, making communication more accessible, engaging, and relevant.
- **Efficient Resource Use:** segmentation can help focus efforts and resources where they will have the most impact, whether in education or building trust.



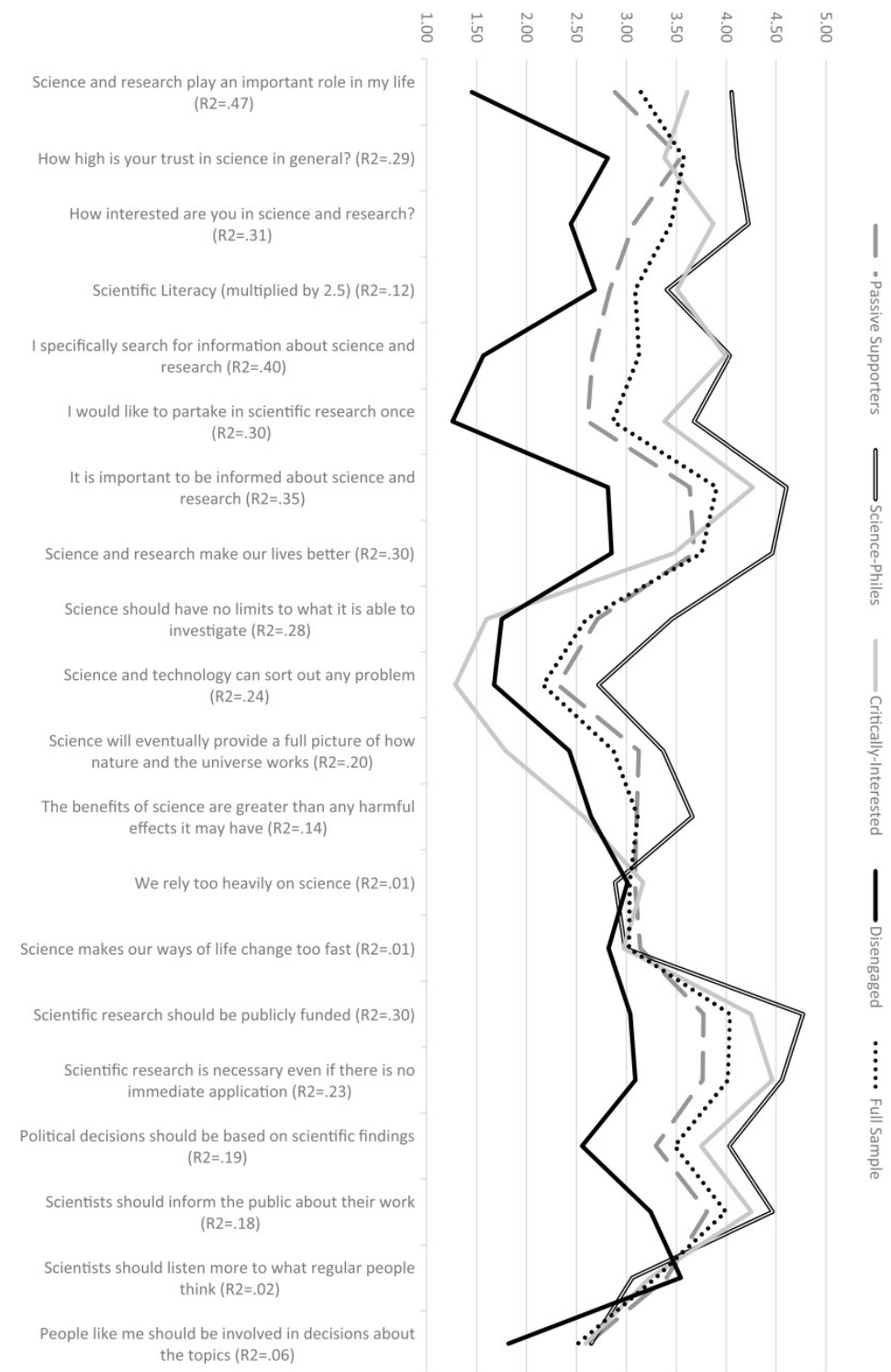
Bertemes, J. P., Haan, S., & Hans, D. (Eds.). (2024). *50 essentials on science communication*. De Gruyter.
<https://doi.org/10.1515/9783110763577>

Public segmentation

Methods

- **Survey:** Science Barometer Switzerland (2016) with 1051 respondents.
- **Segmentation:** Latent class analysis (LCA) based on 20 items covering various dimensions: cognitive (knowledge), affective (trust), and conative (actions) aspects of attitudes towards science.
- **Media use:** Analysis of media use patterns (traditional and online) and engagement with scientific content.

Schäfer, M. S., Fuchslin, T., Metag, J., Kristiansen, S., & Rauchfleisch, A. (2018). The different audiences of science communication: A segmentation analysis of the Swiss population's perceptions of science and their information and media use patterns. *Public Understanding of Science*, 27(7), 836–856.
<https://doi.org/10.1177/0963662517752886>



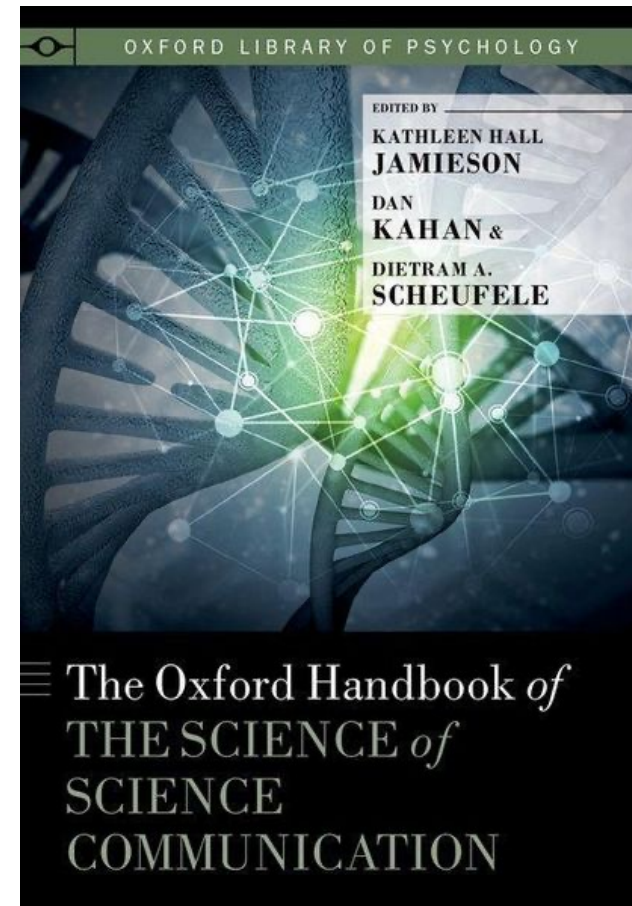
Public Segmentation

Group	Traits	Science Views	Demographics
Sciencephiles (ca. 30%)	- Highest literacy, interest, and trust in science. - Actively seek info. - Optimistic about science's role in society.	- Strong supporters. - Believe in public funding and science's ability to improve lives.	- Highest education. - Heavy use of Internet, media, museums, books.
Critically interested (ca. 15%)	- High knowledge, but critical. - Support research with limits. - Liberal, religious.	- Support science but favor constraints. - Want public funding and political use of science.	- Highly educated. - Religious. - Moderate media use, critical of coverage.
Passive supporters (ca. 40%)	- Moderate interest and trust. - Rarely seek info. - Support public funding with limits.	- Support science but with reservations. - Favor research funding but not fully engaged.	- Moderate education. - Moderate media use, mostly newspapers. - Less engaged.
Disengaged (ca. 15%)	- Lowest knowledge, interest, and trust. - Skeptical of science's impact. - Rarely engage with scientific topics	- Least supportive, - Favor research limits. - Skeptical of science's societal benefits.	- Lowest education. - TV and radio are main sources. - Least engaged.

Public Segmentation

“Science communication scholars are publishing more and more segmentation analyses as they further our understanding of different audiences and their characteristics. They follow different aims, are therefore difficult to compare and do not lend themselves to more generalisable and theoretical knowledge production.”

Füchslin, T. (2019). Science communication scholars use more and more segmentation analyses: Can we take them to the next level? *Public Understanding of Science*, 28(7), 854-864.
<https://doi.org/10.1177/0963662519850086>



Science communication has itself become a target of study: The "science of science communication" is today an interdisciplinary field that studies how various dimensions of science information. The focus on objective, measurable outcomes in this literature has led to increased calls for evaluation of science communication.

HOW SHOULD WE EVALUATE SCIENCE COMMUNICATION?

**Think of a science communication effort and
consider how you would evaluate its impact**



Evaluation of science communication

Stage	INPUTS	OUTPUTS	OUTCOMES	IMPACTS
Focus	What financial and personnel resources are invested in the science communication project?	<i>Primary:</i> What kind of and how many outreach activities are created? What kind of and how many online channels and marketing measures are used? <i>Secondary:</i> What online reach and media coverage are achieved?	<i>Direct:</i> Which audiences participate and how to they respond to the activities? <i>Indirect:</i> What cognitive, emotional, attitudinal, or behavioral effects on audiences are realized?	What are the substantial, long-term values of the science communication project for society and science?
Indicator	Financial: Funding amount, Duration Personnel: Employees, communication experts, project partners	Primary: Activities, online and marketing measures Secondary: Online reach, media coverage	Direct: Participant count, feedback, engagement, fans / followers Indirect: Cognitions, emotions, attitudes, behavior	Society: Societal, educational, environmental, political Science: Publications, awards, follow-up grants
Object	PROJECT	ACTIVITIES MEDIA	AUDIENCES	SOCIETY SCIENCE

Figure 1. Conceptual model for evaluation of science communication projects.

Volk, S. C. (2024). Assessing the Outputs, Outcomes, and Impacts of Science Communication: A Quantitative Content Analysis of 128 Science Communication Projects. *Science Communication*, 10755470241253858. <https://doi.org/10.1177/10755470241253858>

- Study: quantitative content analysis of 128 science communication projects funded by the Swiss National Science Foundation (Agora), 2012-2022.
- Main activities: exhibitions and installations (~40%), workshops and lectures (18%), online platforms (10%), teaching material, apps, and games.
- Online channels: websites (82%), Facebook (36%), YouTube (17%), Twitter/X (16%) - but channel use is reported far more often than channel performance.

Table 2. Primary Outputs of Science Communication Projects (N = 128).

Item	Operationalization	%
ACTIVITIES		
Main activities ^a	Exhibition, installation	39.8
	Workshop, lectures	18.0
	Online platform	10.2
	Learning/teaching material	7.0
	App	4.7
	Science performance, show	4.7
	Film, video, movie	3.1
	Other (e.g., science festival, MOOC, podcast, game)	12.8
COMMUNICATION		
Online communication channels ^a	Website	82.0
	Facebook	35.9
	YouTube	17.2
	Twitter/X	16.4
	Instagram	7.0
	Other/unspecified social media channel	16.4
	Not reported	12.5
Marketing measures ^a	Promotion through network of partners	57.8
	Public poster, flyer, billboard	43.8
	Media relations	42.2
	Newsletter, direct mailing	38.3
	Advertisement (e.g., TV and radio)	13.3
	Not reported	23.4

^aMultiple answers were possible.

- Most projects report some evaluation - but for about one-third, no evaluation is reported.
- Designs are mostly one-off and cross-sectional; pre-post designs are rare (~9%) - and pre-post change alone does not establish causation (no comparison group).
- Methods: feedback forms, (semi-)standardized surveys, interviews, observations - much rests on participants' self-reports.

Table 1. Evaluation Type, Design, and Methods of Science Communication Projects (*N* = 128).

Item	Operationalization	%
Evaluation	Reported	68.7
	Not reported	31.3
Type of evaluation	Summative (ex-post)	53.1
	Pre- and post-test-design	8.6
	Processual evaluation (continuous)	7.0
	Formative (ex-ante)	0
	Not applicable	31.3
Evaluation design	Mainly qualitative (semi-standardized)	28.9
	Mainly quantitative (standardized)	25.8
	Mixed (qualitative and quantitative)	13.3
	Unclear/not applicable	32.0
Evaluation methods ^a	Feedback methods (e.g., guestbook)	42.2
	Standardized surveys	35.2
	User research (e.g., of data collected through apps)	7.8
	Observations	6.3
	Knowledge tests	5.5
	Semi-structured interviews	5.5
	Experiments	0
	Other	0.8
	Unclear/not applicable	31.3

^aMultiple answers were possible.

- Almost half of the projects do not report indirect outcomes
- Metrics for online engagement and reach are inconsistently tracked, making comparisons between projects difficult.

Table 3. Secondary Outputs of Science Communication Projects (N = 128).

Item	Operationalization	%
Coverage in the media	More than 30 reports	11.1
	20 to 29 reports	7.2
	10 to 19 reports	13.2
	1 to 9 reports	44.6
	Not reported	24.2
Reach of online channels	Reported	38.3
	Not reported	49.2
	Not applicable	12.5

“a systematic assessment of the effectiveness of these activities is rare, as few projects apply rigorous evaluation designs and combine multiple evaluation methods. Furthermore, many projects emphasize media attention and participant count, but neglect reporting on the effects on audiences and societal impact.”

Summary

- **Historical perspective:** Science communication has evolved from elite scholarly exchanges to mass media and digital platforms, alongside shifts in communication models: from the **deficit** model (one-way transmission of knowledge), to the **dialogue** model (two-way interaction), and the **participatory** model (co-creation with the public). This reflects a growing emphasis on engagement and public involvement in science.
- **Stakeholders and public segmentation:** Effective science communication requires understanding and addressing the needs of diverse groups, from scientists to the public, to ensure mutual understanding and informed decision-making. Segmenting audiences may allow communicators to tailor messages, ensuring that information resonates with specific groups, but current segmentation efforts may lack generality.
- **Science of SCICOM and its evaluation:** Evaluating the impact of science communication helps refine strategies, ensuring that efforts are not just visible but efficacious in fostering public understanding and engagement. Current surveys suggest that current evaluation efforts are suboptimal.